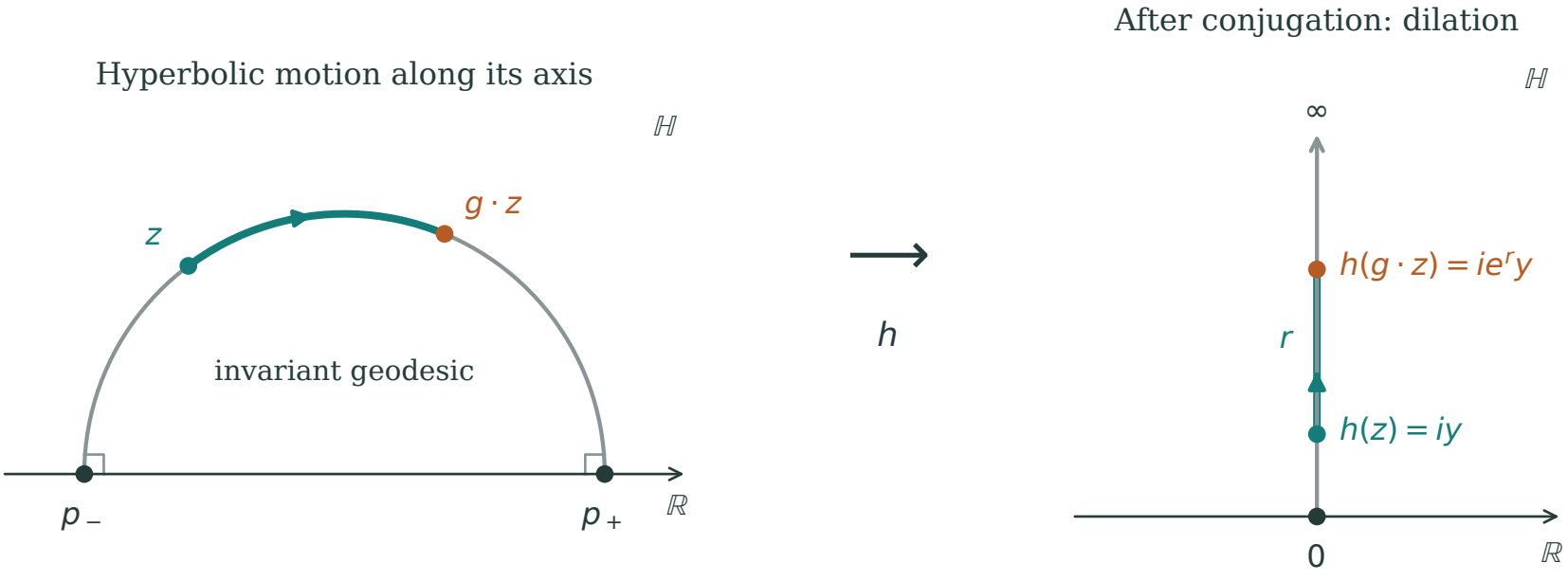
Hyperbolic motion along its axis
$\mathbb{H}$
z
$g \cdot z$
invariant geodesic
p_-
p_+
$\mathbb{R}$
h
After conjugation: dilation
$\mathbb{H}$
∞
$h(g \cdot z) = ie^r y$
r
$h(z) = iy$
0
$\mathbb{R}$